

To: EGR Corporation
 From: Fia Villacorta, Daniel Wang, Shaine Wei
 Subject: Design Challenge 4 - SportBot-RubeGoldberg Assignment
 Date: 15 April 2026

Performance Summary Sketches

SportBot Action Summary

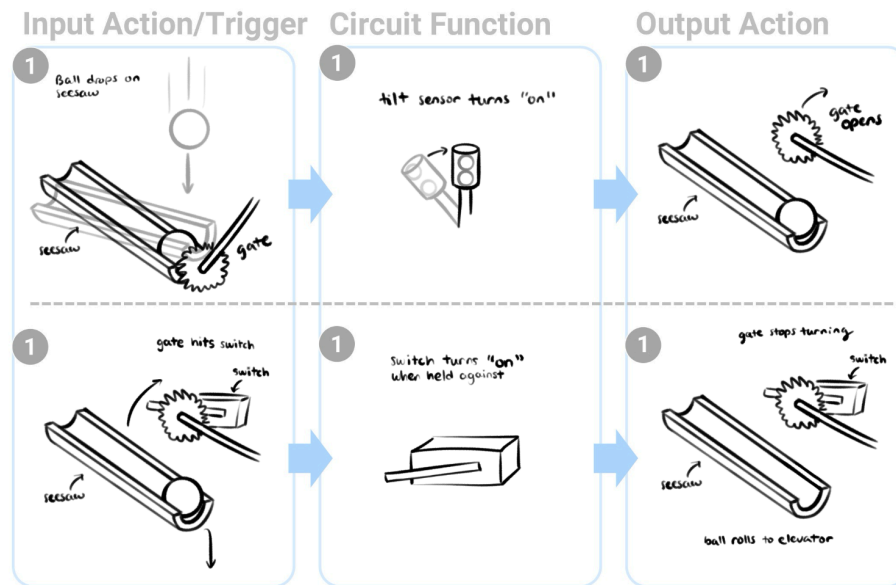


Figure 1. Electronics Circuit SportBot Action Summary Sketch

SportBot Action Summary

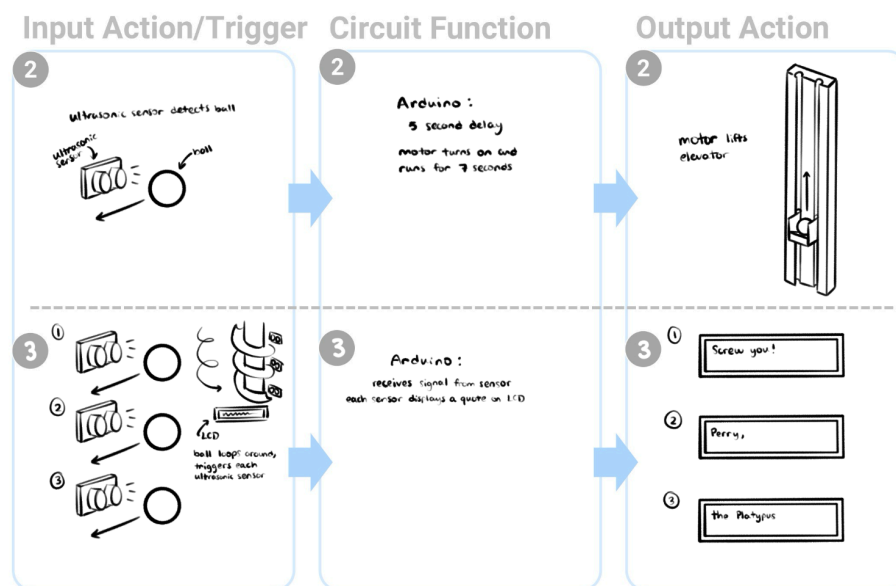
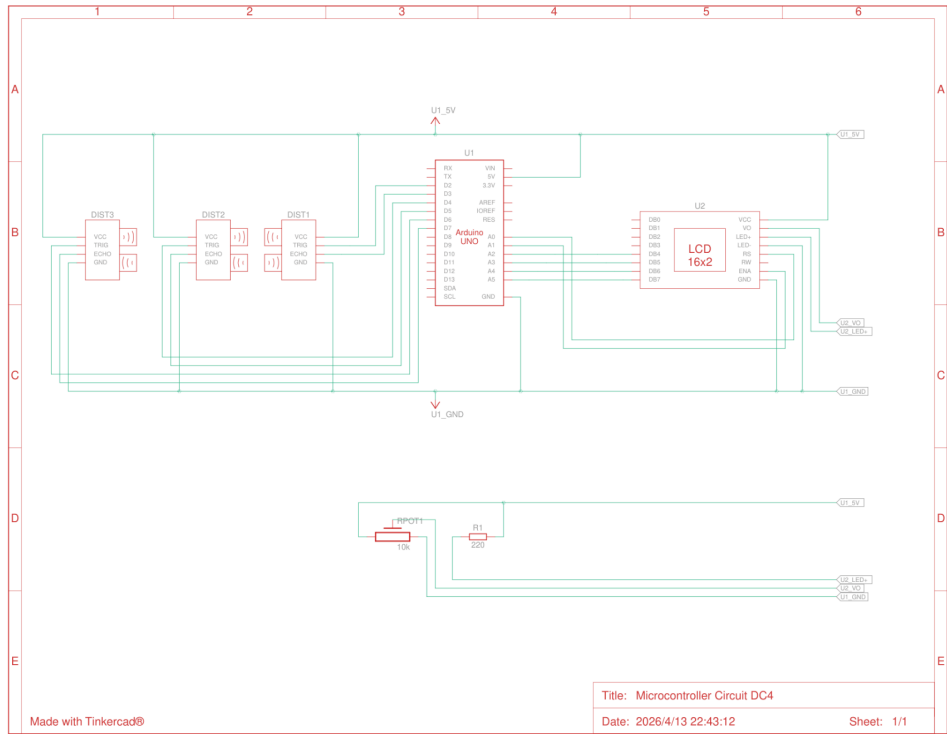
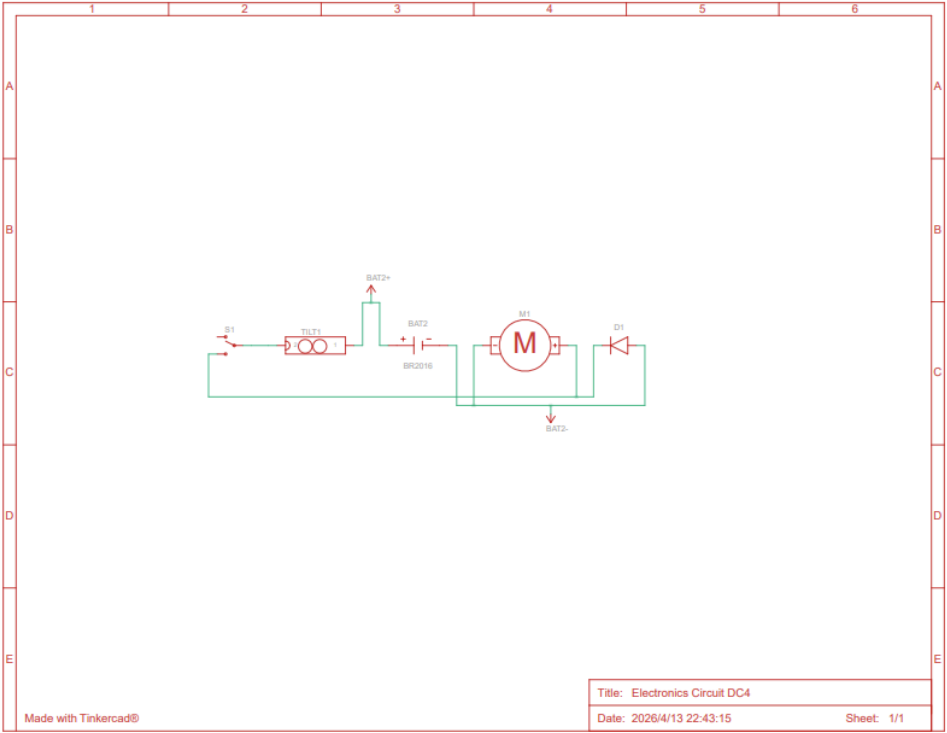


Figure 2. Elevator and Microcontroller Circuit SportBot Action Summary Sketch

Circuit Schematics



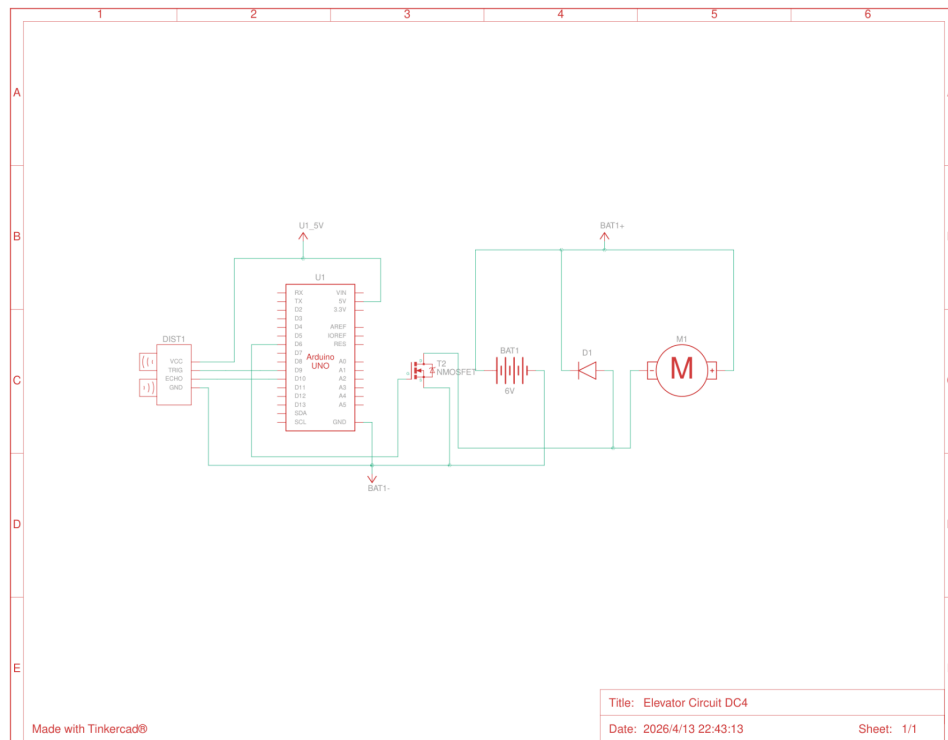


Figure 5. Microcontroller Circuit Schematic

Green Design Element Description

The team used a mix of 3D-printed parts, wood, and electronics that was efficient in material use and modular construction. The wood is considered a renewable material and the 3D-printed parts were custom-made which meant that we only used as much material that was needed, leading to minimal waste. Many of our structural components were taken apart to be used in the future. Additionally, a lot of the components that the team used for the electronics and circuits were also reusable and put back into the materials in the Foundry for other students to use.